Week 3

Course Level Objectives:

- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Construct frequency and relative frequency tables for a categorical variable. (CLO 2) **2.** Compute a sample proportion $\hat{p}$. (CLO 2) **3.** Distinguish the parameter p from the statistic $\hat{p}$. (CLO 2) **4.** Compute overall and conditional proportions from a two-way table, choosing the correct denominator. (CLO 2) **5.** Compute and interpret a difference in proportions. (CLO 2) **6.** Interpret bar charts, pie charts, side-by-side bar charts, and segmented bar charts. (CLO 2)	e-Text: 2.1 Categorical Variables Interactive Lecture: 2.1 Categorical Variables Slides: 2.1 Categorical Variables	Participation: Frequency and Relative Frequency Participation: Finding Proportions from Two-Way Tables – Your Turn: Two-Way Table Proportions (pet ownership by housing)	Online Homework: 2.1 Categorical Variables (due Sun Sep 13, 11:59 PM) Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus
7. Describe the shape of a distribution (symmetric, left-skewed, right-skewed) from a dotplot or histogram. (CLO 2) **8.** Compute the mean and median of a dataset. (CLO 2) **9.** Explain how skewness and outliers affect the mean and the median (resistance). (CLO 2) **10.** Interpret the standard deviation as the typical distance from the mean. (CLO 2) **11.** Apply the 95% rule to estimate the mean and standard deviation from a bell-shaped dotplot. (CLO 2)	e-Text: 2.2 One Quantitative Variable Interactive Lecture: 2.2 One Quantitative Variable Slides: 2.2 One Quantitative Variable	Participation: Quantitative Variables – shape, mean vs. median; Your Turn: Mean vs. Median with an Outlier (commute times) Estimation activity: estimate the mean and SD of a dotplot without calculating (95% rule)	Online Homework: 2.2 One Quantitative Variable (due Sun Sep 13, 11:59 PM) Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus